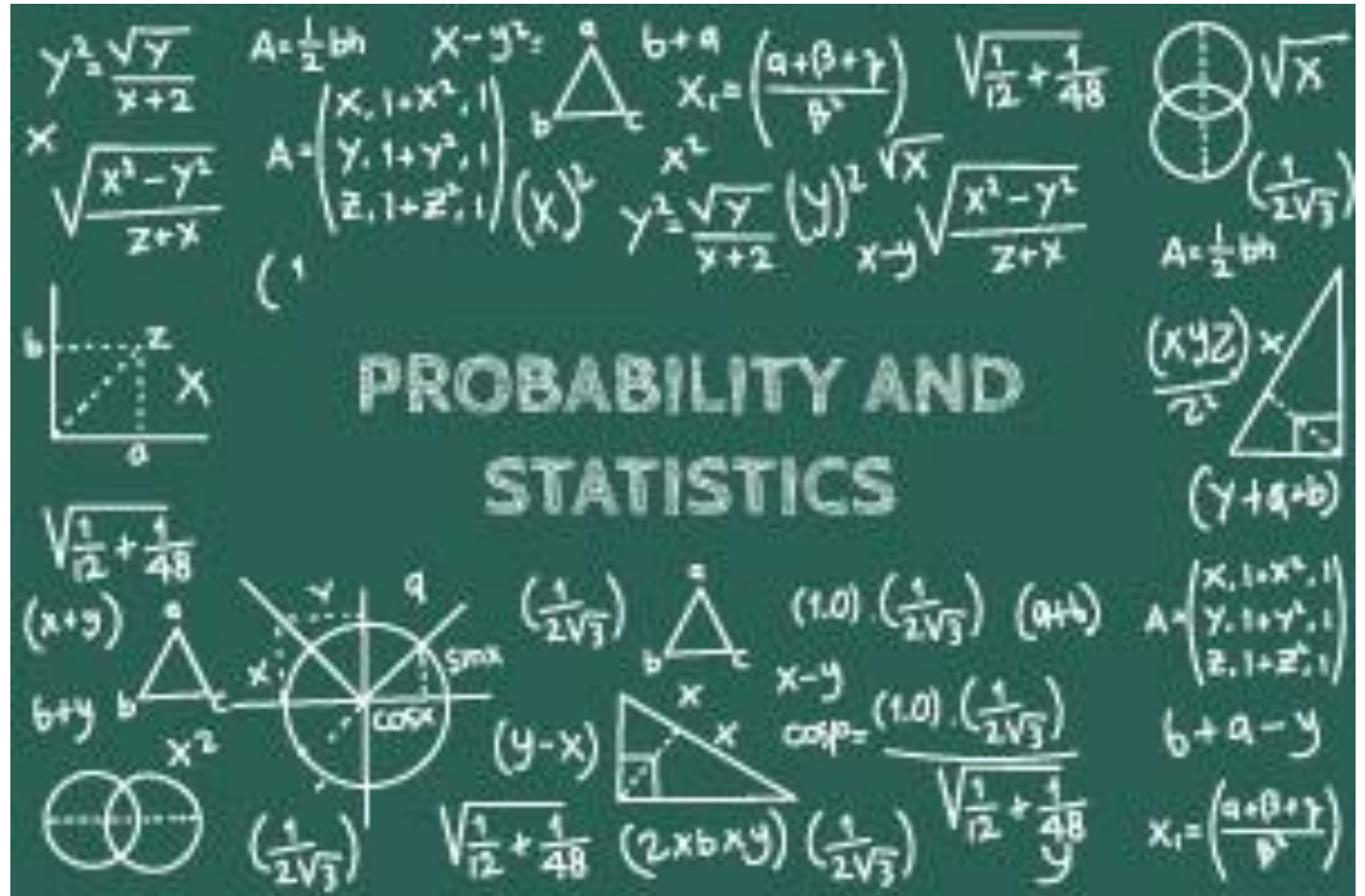


Lec # 04
ES111

Counting Sample Points

Book Reading: Sec. 2.3

@M. Tahir Naseem



Discussion Plan

1. Combinations

1. Permutations

3. Practice Problems (Sec. 2.3)



To calculate probability

To calculate probability

1. calculate "sample space"

To calculate probability

1. calculate "sample space"
2. Identify "event" of interest

To calculate probability

1. calculate "sample space"
2. Identify "event" of interest
3. Divide "event" by "sample space" $\longrightarrow$ probability

To calculate probability

1. calculate "sample space"  Today's class
2. Identify "event" of interest
3. Divide "event" by "sample space"  probability

Sample Space

Sample Space



A set of all possible outcomes of statistical experiment is called sample space

Counting Sample points

Counting Sample points

"How many different ways we can order?"



Counting Sample points

"How many different ways we can order?"



2

x

1

Counting Sample points

"How many different ways we can order?"



X

1

Counting Sample points

"How many different ways we can order?"



3

X

1

Counting Sample points

"How many different ways we can order?"



3

X

Counting Sample points

"How many different ways we can order?"



3

X

2

Counting Sample points

"How many different ways we can order?"



3

X



2

X



Counting Sample points

"How many different ways we can order?"



3

x



2

x



4

Counting Sample points

Multiplication Rule



3

X



2

X



4

Counting Sample points

Multiplication Rule

Categories



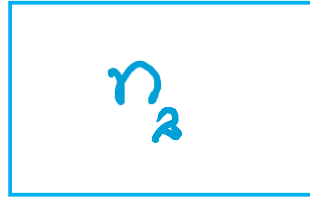
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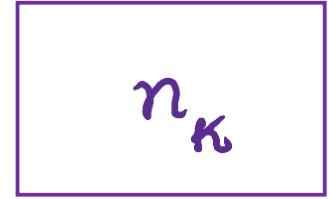
Counting Sample points

Multiplication Rule

Categories



...



All possibilities = $n_1 \times n_2 \cdots n_k$



How many sample points are there in the sample space when a pair of dice is thrown once?



How many sample points are there in the sample space when a **pair of dice** is thrown once?





How many sample points are there in the sample space when a **pair of dice** is thrown once?



$$n_1 = 6$$



$$n_2 = 6$$



How many sample points are there in the sample space when a **pair of dice** is thrown once?



$$n_1 = 6$$



$$n_2 = 6$$

$$= n_1 \times n_2 = 6 \times 6$$



Sam is going to assemble a computer by himself. He has the choice of **chips from two brands**, a **hard drive from four**, **memory from three**, and **an accessory bundle from five local stores**. How many different ways can Sam order the parts?



Sam is going to assemble a computer by himself. He has the choice of **chips from two brands**, a **hard drive from four**, **memory from three**, and **an accessory bundle from five local stores**. How many different ways can Sam order the parts?

$$= 2 \times 4 \times 3 \times 5 = 120$$

Ans



A certain brand of shoes comes in 5 different styles, with each style available in 4 distinct colors. If the store wishes to display pairs of these shoes showing all of its various styles and colors, how many different pairs will the store have on display?



A certain brand of shoes comes in **5 different styles**, with each style available **in 4 distinct colors**. If the store wishes to display pairs of these shoes showing all of its various styles and colors, how many different pairs will the store have on display?

$$= 5 \times 4 = 20$$

Ans

Counting Sample points

Multiplication Rule
↓

Repetition of Events



In how many different ways can a true-false test consisting of 9 questions be answered?



In how many different ways can a **true-false** test consisting of **9 questions** be answered?



In how many different ways can a true-false test consisting of 9 questions be answered?

$$2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2$$



In how many different ways can a **true-false** test consisting of **9 questions** be answered?

$$2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 2^9$$



If a multiple-choice test consists of 5 questions, each with 4 possible answers of which only 1 is correct, in how many different ways can a student check off one answer to each question?



If a multiple-choice test consists of **5 questions**, each **with 4 possible answers** of which only 1 is correct, in how many different ways can a student check off one answer to each question?

$$= 4^5$$

Ans

Counting Sample points



1. Multiplication Rule

$$= n_1 \times n_2 \times \dots \times n_k$$

2. Combinations

Combinations

Combinations → Arrangements in which order does not matter.

Combinations → Arrangements in which order does not matter.
Arrangement in which order matters → permutations

Combinations

→ Arrangements in which **order** does not matter.

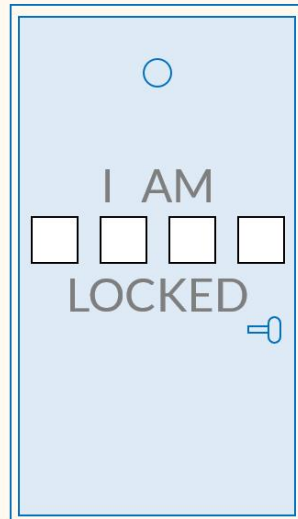
Arrangement in which **order matters** → **permutations**

Confused

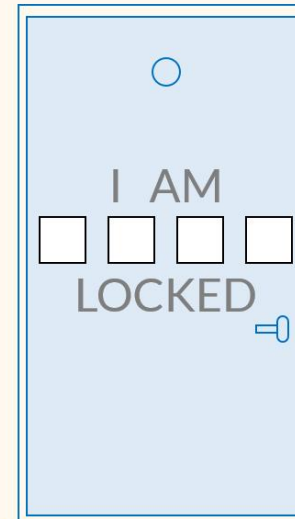


Should I unlock with Permutation or Combination ?

Permutation



Combination

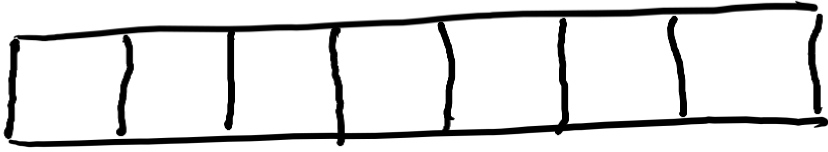


Combinations → Arrangements in which order does not matter.

A B C same B A C

Combinations

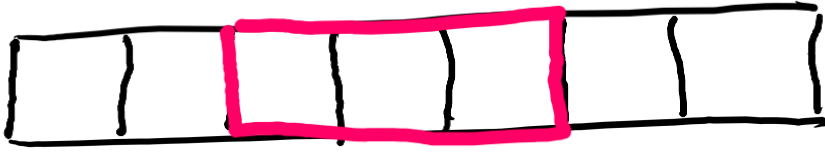
n outcomes



Combinations

γ outcomes

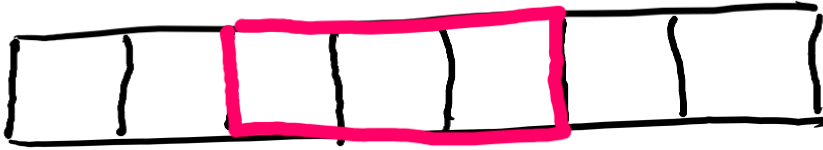
n outcomes



Combinations

r outcomes

n outcomes

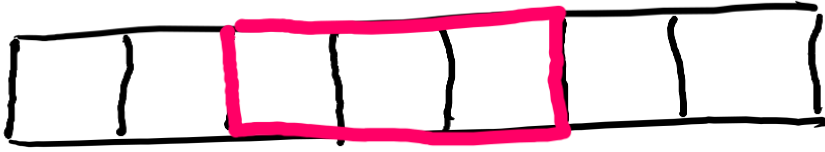


$$\text{combination} = \frac{n!}{r! (n-r)!}$$

Combinations

r outcomes

n outcomes

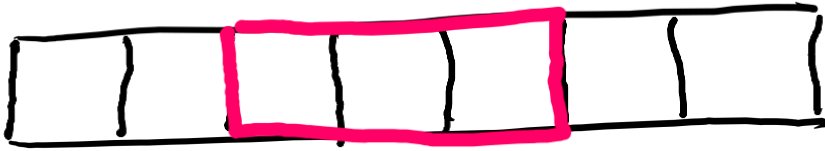


$$\text{combination} = \frac{n!}{r! (n-r)!} = {}^n C_r$$

Combinations

r outcomes

n outcomes



$$\text{combination} = \frac{n!}{r! (n-r)!} = {}^nC_r = \binom{n}{r}$$

Counting Sample points



1. Multiplication Rule

$$= n_1 \times n_2 \times \dots \times n_k$$

2. Combinations

2.1 "no constraints"

Counting Sample points



1. Multiplication Rule

$$= n_1 \times n_2 \times \dots \times n_k$$

2. Combinations

2.1 "no constraints"

$${}_n C_r = \frac{n!}{r! (n-r)!}$$



In one year, three same awards will be given to a class of 25 graduate students in a statistics department. If each student can receive at most one award, how many possible selections are there?



In one year, **three same awards** will be given to a class of **25 graduate** students in a statistics department. If each student can receive at most one award, **how many possible** selections are there?



In one year, **three same awards** will be given to a class of **25 graduate** students in a statistics department. If each student can receive at most one award, **how many possible** selections are there?

$${}_nC_r = \frac{n!}{r! (n-r)!}$$



In one year, **three same awards** will be given to a class of **25 graduate** students in a statistics department. If each student can receive at most one award, **how many possible** selections are there?

$$n = 25$$
$$r = 3$$

$${}_nC_r = \frac{n!}{r! (n-r)!} = \frac{25!}{3! (25-3)!} \quad \underline{\text{Ans}}$$



How many ways are there to select 3 candidates from 8 equally qualified recent graduates for openings in an accounting firm?



How many ways are there to **select 3 candidates** from **8 equally qualified** recent graduates for openings in an accounting firm?



How many ways are there to **select 3 candidates** from **8 equally qualified** recent graduates for openings in an accounting firm?

$${}_8C_3 = \frac{8!}{3!(8-3)!}$$

$${}_nC_r = \frac{n!}{r!(n-r)!}$$



How many ways are there to **select 3 candidates** from **8 equally qualified** recent graduates for openings in an accounting firm?

$${}_nC_r = \frac{n!}{r! (n-r)!}$$

$${}_8C_3 = \frac{8!}{3! (8-3)!} = \frac{8!}{3! 5!}$$



How many ways are there to **select 3 candidates** from **8 equally qualified** recent graduates for openings in an accounting firm?

$${}_nC_r = \frac{n!}{r! (n-r)!}$$

$$\begin{aligned} {}_8C_3 &= \frac{8!}{3! (8-3)!} = \frac{8!}{3! 5!} \\ &= \frac{8 \times 7 \times 6 \times 5!}{3! 5!} \end{aligned}$$



How many ways are there to **select 3 candidates** from **8 equally qualified** recent graduates for openings in an accounting firm?

$${}_nC_r = \frac{n!}{r! (n-r)!}$$

$$\begin{aligned} {}_8C_3 &= \frac{8!}{3! (8-3)!} = \frac{8!}{3! 5!} \\ &= \frac{8 \times 7 \times \cancel{6} \times \cancel{5}!}{\cancel{3!} \cancel{5!}} \end{aligned}$$



How many ways are there to **select 3 candidates** from **8 equally qualified** recent graduates for openings in an accounting firm?

$${}_nC_r = \frac{n!}{r! (n-r)!}$$

$$\begin{aligned} {}_8C_3 &= \frac{8!}{3! (8-3)!} = \frac{8!}{3! 5!} \\ &= \frac{8 \times 7 \times \cancel{6} \times \cancel{5}!}{\cancel{3!} \cancel{5!}} = 56 \text{ ans} \end{aligned}$$

Counting Sample points

✓
1. Multiplication Rule

$$= n_1 \times n_2 \times \dots \times n_k$$

2. Combinations

✓
2.1 "no constraints"

$${}_n C_r = \frac{n!}{r! (n-r)!}$$

Counting Sample points

✓ 1. Multiplication Rule

$$= n_1 \times n_2 \times \dots \times n_k$$

2. Combinations

✓ 2.1 "no constraints"

$${}_n C_r = \frac{n!}{r! (n-r)!}$$

2.2 no constraints but "selection is made from different categories"

Combinations

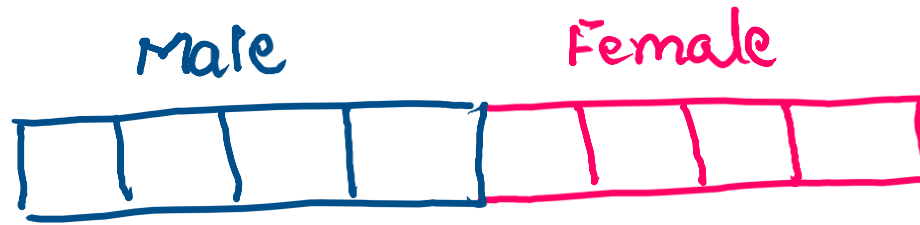
2.2

no constraints but "selection is made from different categories"

Combinations

2.2

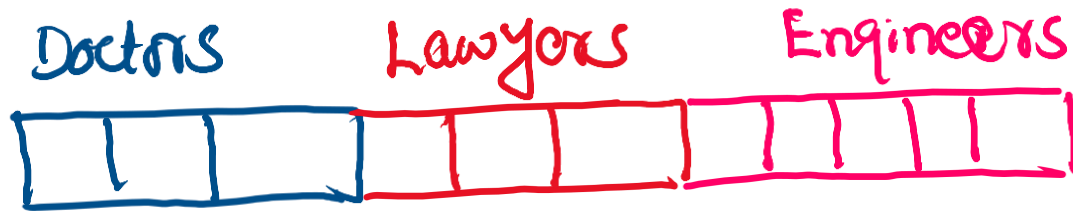
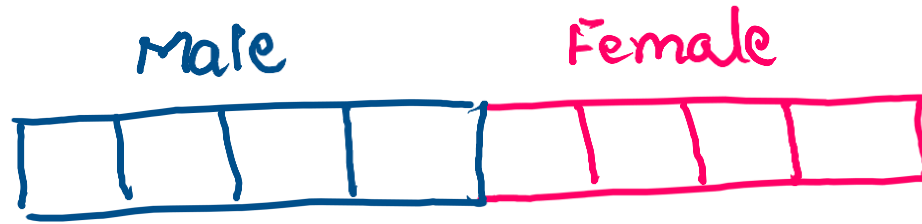
no constraints but "selection is made from different categories"



Combinations

2.2

no constraints but "selection is made from different categories"

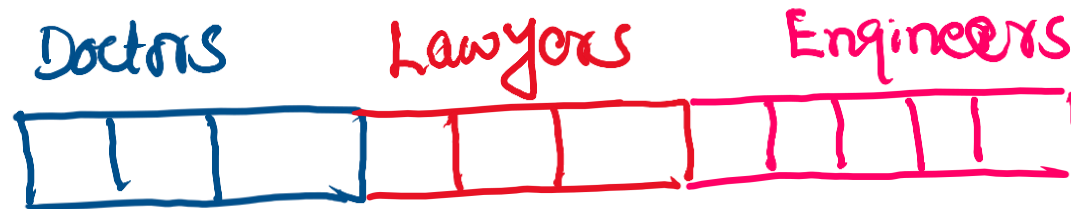
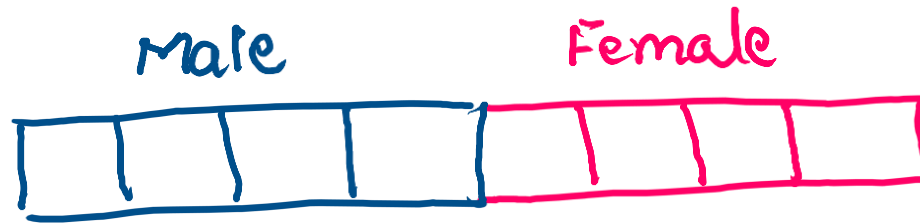


Combinations

2.2

no constraints but "selection is made from different categories"

→ "select a team of 3"

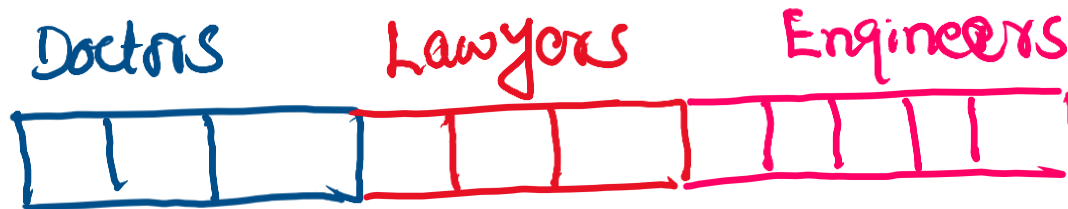
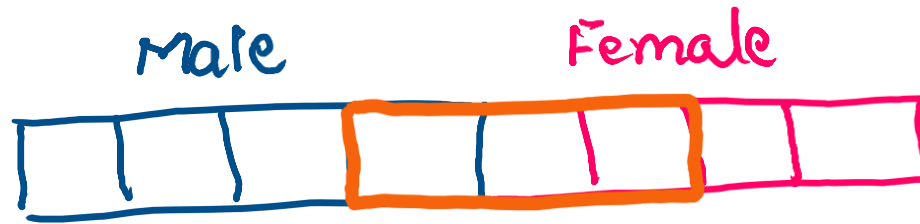


Combinations

2.2

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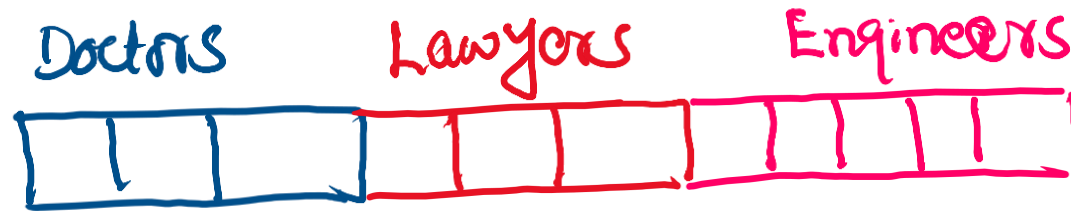
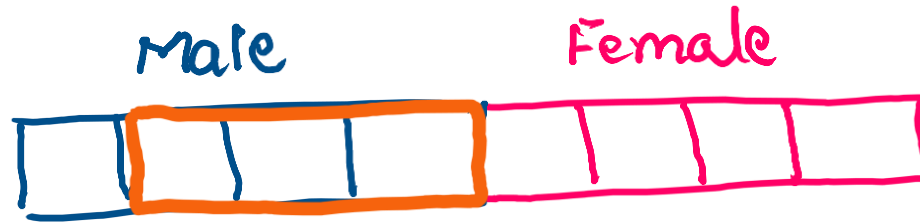


Combinations

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no constraints but "selection is made from different categories"

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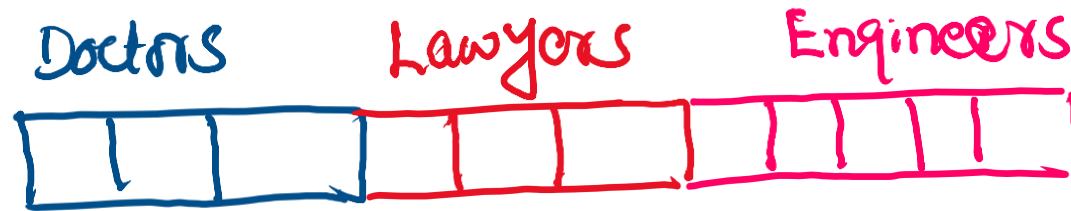
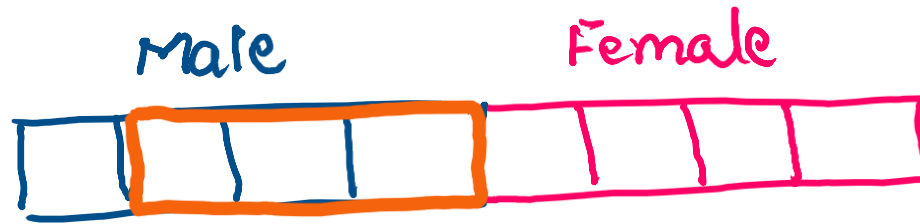


Combinations

2.2

no constraints but "selection is made from different categories"

→ "select a team of 3"



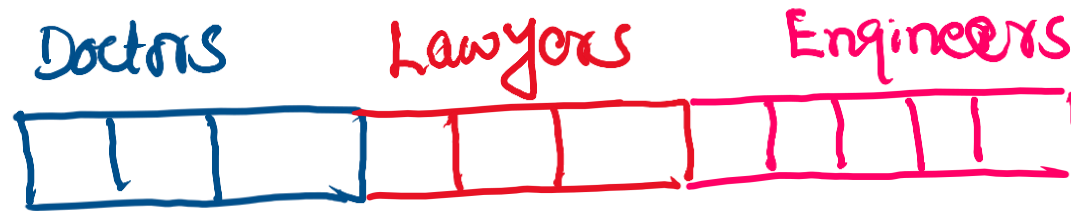
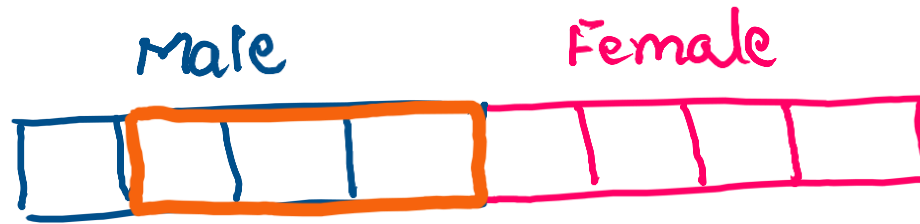
"no constraint over selection"

Combinations

2.2

no constraints but "selection is made from different categories"

→ "select a team of 3"



"no constraint over selection"

$${}_n C_r = \frac{n!}{r! (n-r)!}$$



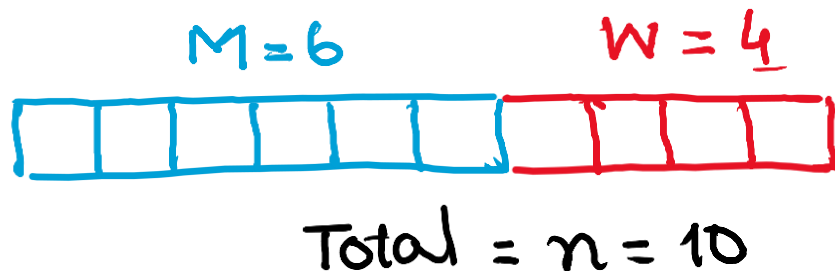
A committee of 5 people is to be formed from a group of 6 men and 4 women. How many ways it is possible to form a committee.



A committee of **5 people** is to be formed from a group of **6 men** and **4 women**. How many ways it is possible to form a committee.



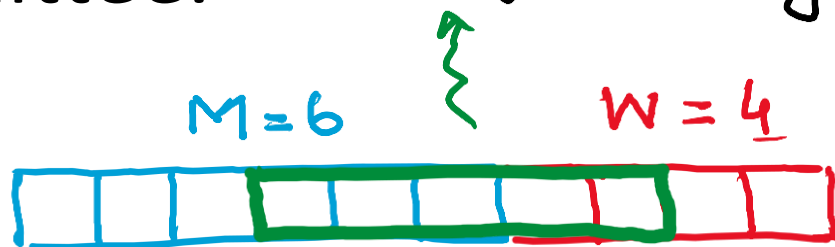
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A committee of **5 people** is to be formed from a group of **6 men** and **4 women**. How many ways it is possible to form a committee.

'can be any 5 shown as example'

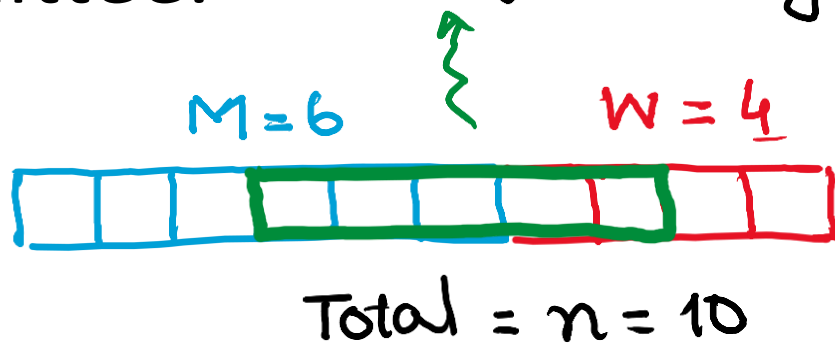


Total = $n = 10$



A committee of **5 people** is to be formed from a group of **6 men** and **4 women**. How many ways it is possible to form a committee.

'can be any 5 shown as example'

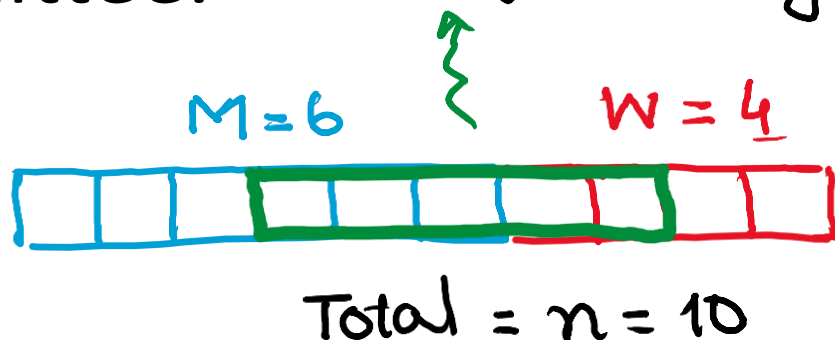


$${}_n C_r = \frac{n!}{r! (n-r)!}$$



A committee of **5 people** is to be formed from a group of **6 men** and **4 women**. How many ways it is possible to form a committee.

'can be any 5 shown as example'



$${}_nC_r = \frac{n!}{r! (n-r)!}$$

$${}_{10}C_5 = \frac{10!}{5! (10-5)!}$$

Ans



If 3 books are picked at random from a shelf containing 5 novels, 3 books of poems, and a dictionary. In how many ways books can be picked.



If **3 books** are picked at random from a shelf containing **5 novels**, **3 books of poems**, and **a dictionary**. In how many ways books can be picked.



If **3 books** are picked at random from a shelf containing **5 novels**, **3 books of poems**, and **a dictionary**. In how many ways books can be picked.

$${}^nC_r = \frac{n!}{r! (n-r)!}$$



If **3 books** are picked at random from a shelf containing **5 novels**, **3 books of poems**, and **a dictionary**. In how many ways books can be picked.

$${}^9C_3 = \frac{9!}{3! (9-3)!}$$

Ans

$${}^nC_r = \frac{n!}{r! (n-r)!}$$

$$n = 9$$

$$r = 3$$

Counting Sample points

✓ 1. Multiplication Rule

$$= n_1 \times n_2 \times \dots \times n_k$$

2. Combinations

✓ 2.1 "no constraints"

$${}_n C_r = \frac{n!}{r! (n-r)!}$$

Same

✓ 2.2 no constraints but "selection is made from different categories"

Counting Sample points

✓ 1. Multiplication Rule

$$= n_1 \times n_2 \times \dots \times n_k$$

2. Combinations

✓ 2.1 "no constraints"

$${}_n C_r = \frac{n!}{r! (n-r)!}$$

Same

✓ 2.2 no constraints but "selection is made from different categories"

2.3 with constraints

Counting Sample points

✓ 1. Multiplication Rule

$$= n_1 \times n_2 \times \dots \times n_k$$

2. Combinations

✓ 2.1 "no constraints"

$${}_n C_r = \frac{n!}{r! (n-r)!}$$

Same

✓ 2.2 no constraints but "selection is made from different categories"

2.3 with constraints → "constraint selection from each different group"

Combinations

2.3

with constraints

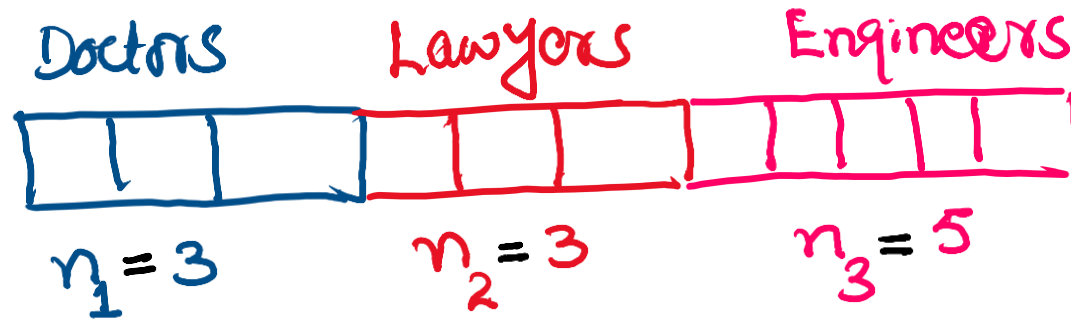
"constraint selection from each different group"

Combinations

(2.3)

with constraints

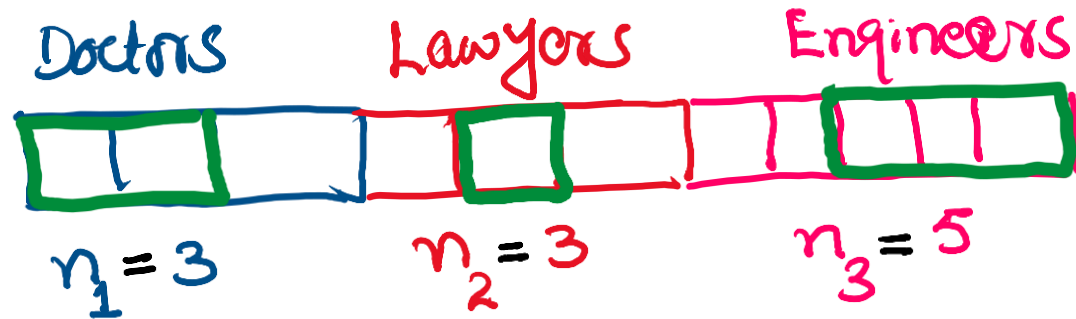
"constraint selection from each different group"



Combinations

(2.3) with constraints $\rightarrow$ "constraint selection from each different group"

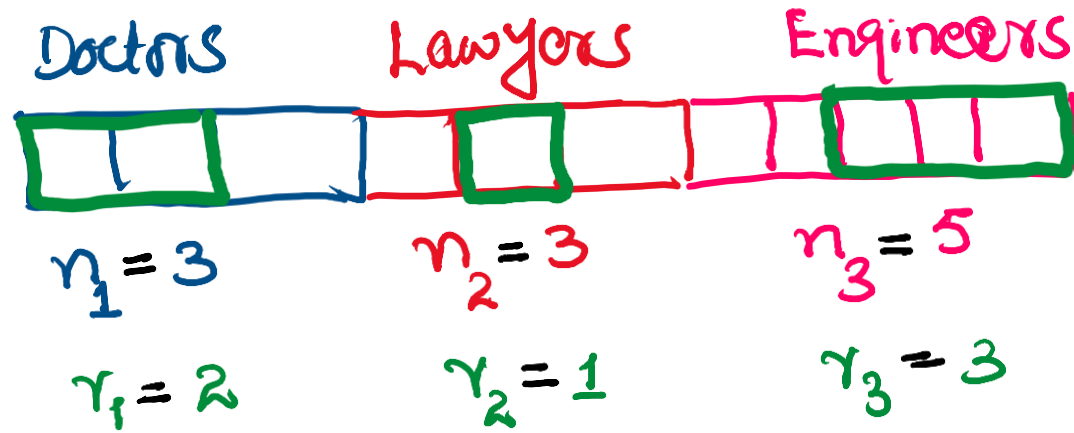
suppose Q "select 2 doctors 1 lawyer and 3 engineers"



Combinations

(2.3) with constraints $\rightarrow$ "constraint selection from each different group"

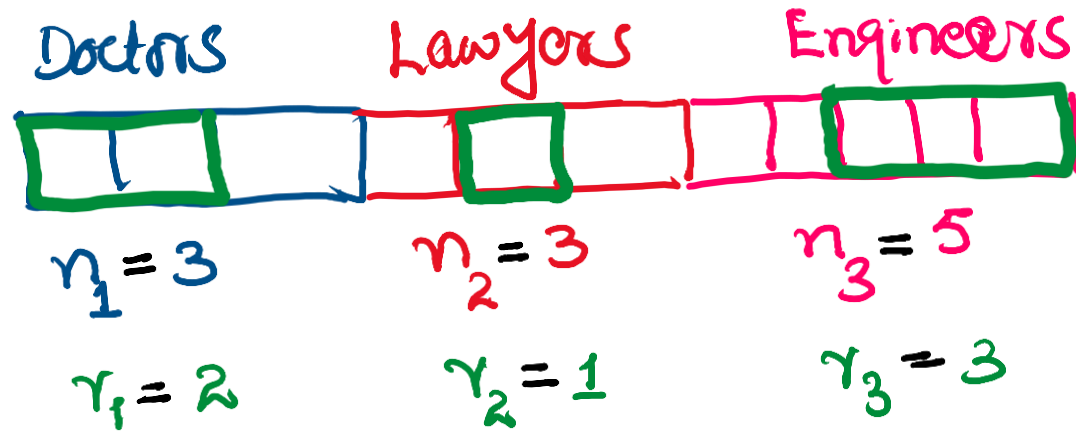
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Combinations

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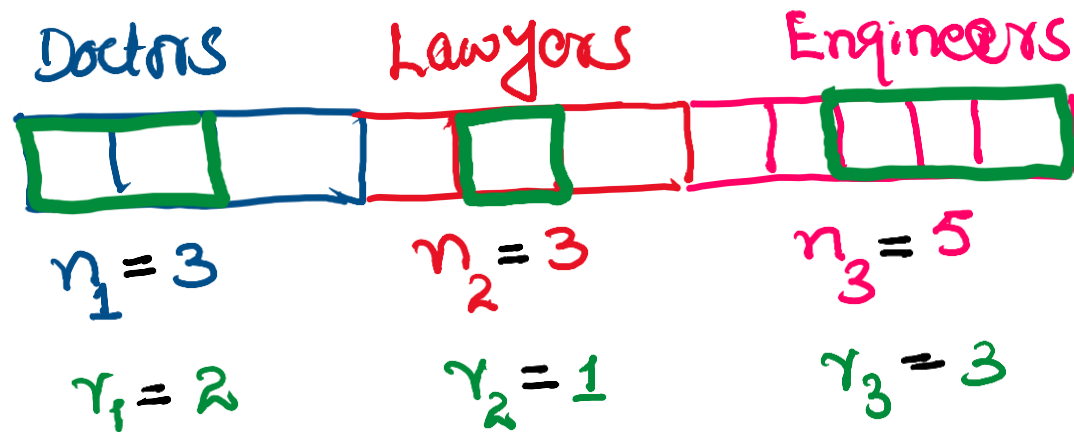


$${}_{n_1}C_{r_1}$$

Combinations

(2.3) with constraints $\rightarrow$ "constraint selection from each different group"

suppose Q "select 2 doctors 1 lawyer and 3 engineers"



$${}_{n_1}C_{r_1}$$

$${}_{n_2}C_{r_2}$$

Combinations

(2.3) with constraints

"constraint selection from each different group"

suppose Q "select 2 doctors 1 lawyer and 3 engineers"



$$n_1 = 3$$

$$r_1 = 2$$

$${}_{n_1}C_{r_1}$$

$$n_2 = 3$$

$$r_2 = 1$$

$${}_{n_2}C_{r_2}$$

$$n_3 = 5$$

$$r_3 = 3$$

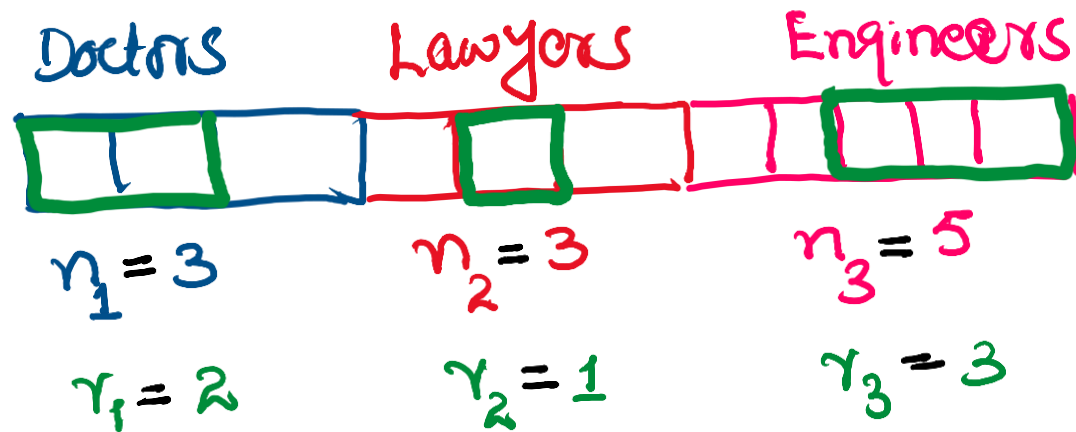
$${}_{n_3}C_{r_3}$$

Combinations

(2.3) with constraints

"constraint selection from each different group"

suppose Q "select 2 doctors 1 lawyer and 3 engineers"



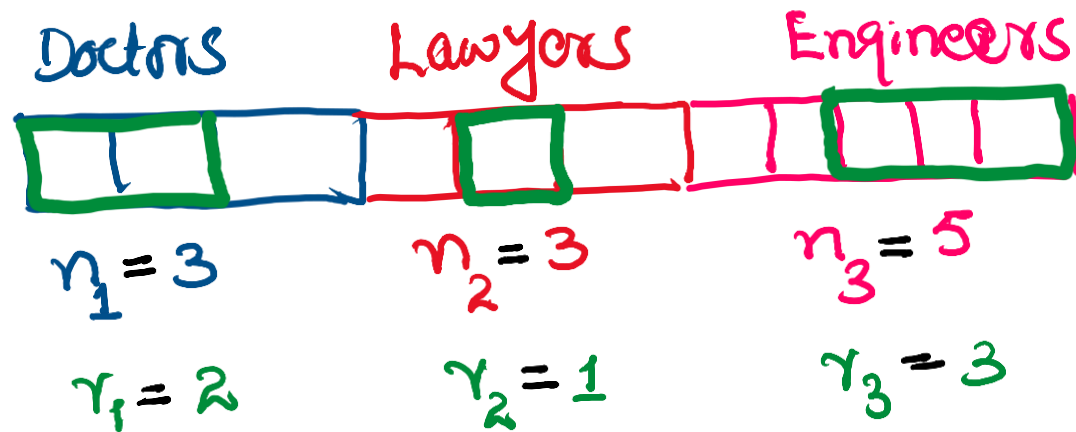
$$\text{Total possibilities} = {}^{n_1}C_{r_1} \times {}^{n_2}C_{r_2} \times {}^{n_3}C_{r_3}$$

Combinations

(2.3) with constraints

"constraint selection from each different group"

suppose Q "select 2 doctors 1 lawyer and 3 engineers"



$$\begin{aligned} \text{Total possibilities} &= {}^{n_1}C_{r_1} \times {}^{n_2}C_{r_2} \times {}^{n_3}C_{r_3} \\ &= {}^3C_2 \times {}^3C_1 \times {}^5C_3 \end{aligned}$$



A committee of **5 people** is to be formed from a group of **6 men** and **4 women**. How many ways it is possible to form a committee.

we have solved earlier
notice no constraint!

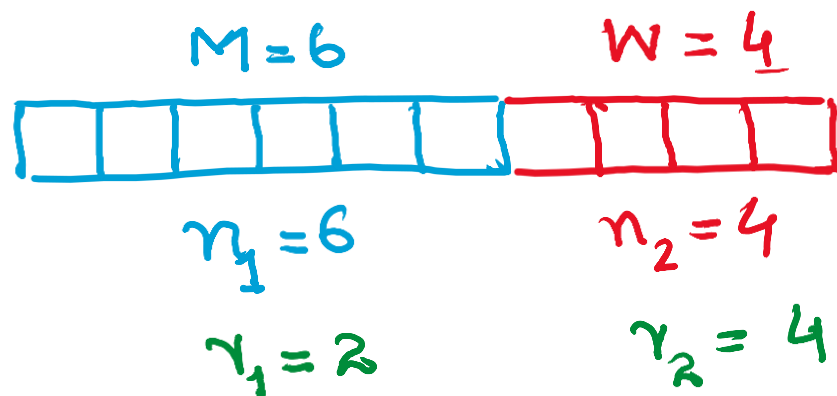


A committee of **5 people** is to be formed from a group of **6 men** and **4 women**. How many ways it is possible to form a committee if there are 2 men and 4 women.

→ "constraint added"



A committee of **5 people** is to be formed from a group of **6 men** and **4 women**. How many ways it is possible to form a committee if there are 2 men and 4 women.

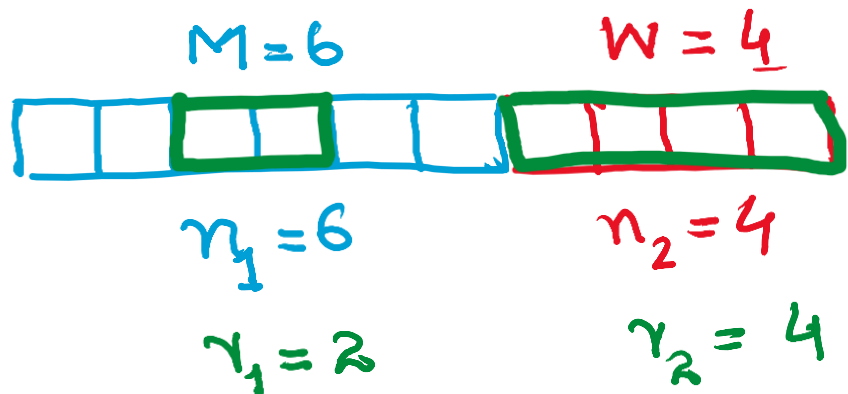


→ "constraint added"



A committee of **5 people** is to be formed from a group of **6 men** and **4 women**. How many ways it is possible to form a committee if there are 2 men and 4 women.

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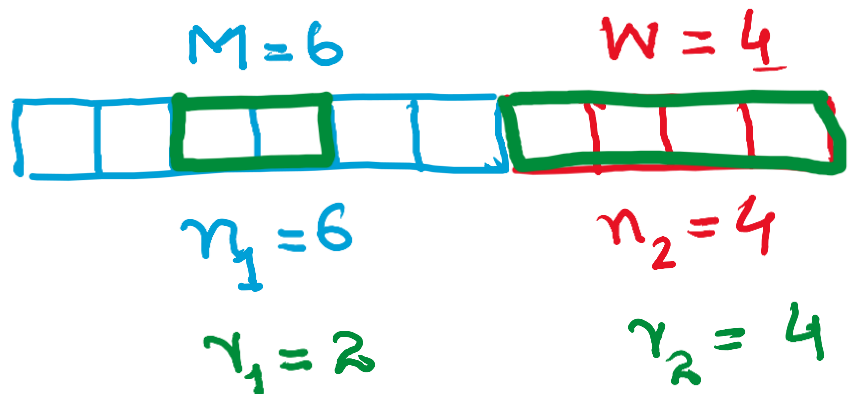


$$= {}^{n_1}C_{r_1} \times {}^{n_2}C_{r_2}$$



A committee of **5 people** is to be formed from a group of **6 men** and **4 women**. How many ways it is possible to form a committee if there are 2 men and 4 women.

→ "constraint added"



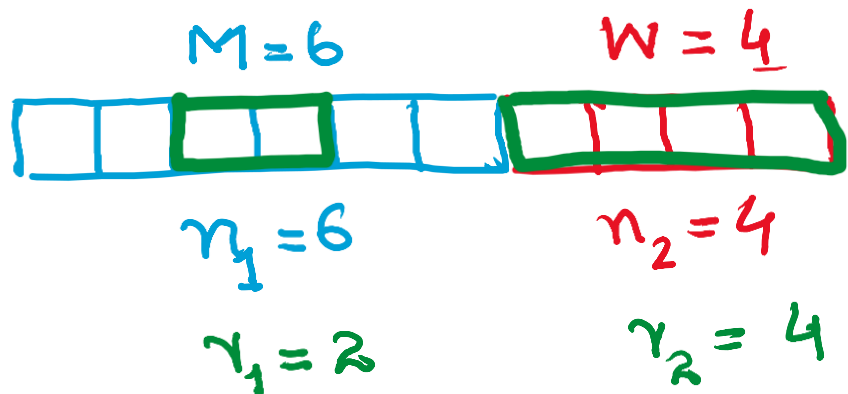
$$= {}^{n_1}_{r_1} C_{r_1} \times {}^{n_2}_{r_2} C_{r_2}$$

$$= {}^6_2 C_2 \times {}^4_4 C_4$$



A committee of **5 people** is to be formed from a group of **6 men** and **4 women**. How many ways it is possible to form a committee if there are 2 men and 4 women.

→ "constraint added"



$$= {}_{n_1}C_{r_1} \times {}_{n_2}C_{r_2}$$

$$= {}_6C_2 \times {}_4C_4$$

$$= {}_6C_2 \times 1 \quad \underline{\underline{\text{Ans}}}$$



From **4 red**, **5 green**, and **6 yellow apples**, how many selections of **9 apples** are possible if 3 of each color are to be selected?

3 constraint



From **4 red**, **5 green**, and **6 yellow apples**, how many selections of **9 apples** are possible if 3 of each color are to be selected?

3 constraint

$$= {}^C_4 \times {}^C_5 \times {}^C_6$$



From **4 red**, **5 green**, and **6 yellow apples**, how many selections of **9 apples** are possible if 3 of each color are to be selected?

3 constraint

$$= {}^4C_3 \times {}^5C_3 \times {}^6C_3$$

Ans

Counting Sample points

✓ 1. Multiplication Rule

$$= n_1 \times n_2 \times \dots \times n_k$$

2. Combinations

✓ 2.1 "no constraints"

$${}_n C_r = \frac{n!}{r! (n-r)!}$$

Same

✓ 2.2 no constraints but "selection is made from different categories"

✓ 2.3 with constraints → "constraint selection from each different group"

$$= {}_{n_1} C_{r_1} \times {}_{n_2} C_{r_2} \times {}_{n_3} C_{r_3} \dots$$

Counting Sample points

✓ 1. Multiplication Rule

$$= n_1 \times n_2 \times \dots \times n_k$$

✓ 2. Combinations

$${}_n C_r = \frac{n!}{r! (n-r)!}$$

Counting Sample points

✓ 1. Multiplication Rule

$$= n_1 \times n_2 \times \dots \times n_k$$

✓ 2. Combinations

$${}_n C_r = \frac{n!}{r! (n-r)!}$$

3. Permutations

Counting Sample points

✓ 1. Multiplication Rule

$$= n_1 \times n_2 \times \dots \times n_k$$

✓ 2. Combinations

$${}_n C_r = \frac{n!}{r! (n-r)!}$$

3. Permutations

3.1 Permutations without constraints

Permutations → Arrangements in which order matters

Permutations → Arrangements in which order matters

$${}_nP_r =$$

Permutations → Arrangements in which order matters

$${}_nP_r = \frac{n!}{(n-r)!}$$

Permutations → Arrangements in which order matters

$${}_nP_r = \frac{n!}{(n-r)!} > {}_nC_r = \frac{n!}{\underbrace{r!}_{\text{extra}} (n-r)!}$$



In how many ways can 5 starting positions on a basketball team be filled with 8 men who can play any of the positions?



In how many ways can **5 starting positions** on a basketball team be filled with **8 men** who can play any of the positions?



In how many ways can **5 starting positions** on a basketball team be filled with **8 men** who can play any of the positions?



In how many ways can **5 starting positions** on a basketball team be filled with **8 men** who can play any of the positions?

$${}_nP_r = \frac{n!}{(n-r)!}$$



In how many ways can **5 starting positions** on a basketball team be filled with **8 men** who can play any of the positions?

$${}_8P_5 = \frac{8!}{(8-5)!} = 6,720$$

Ans

$${}_nP_r = \frac{n!}{(n-r)!}$$



Find the number of ways that 6 teachers can be assigned to 4 sections of an introductory psychology course if no teacher is assigned to more than one section.



Find the number of ways that **6 teachers** can be assigned to **4 sections of an introductory psychology course** if no teacher is assigned to more than one section.



Find the number of ways that **6 teachers** can be assigned to **4 sections of an introductory psychology course** if no teacher is assigned to more than one section.

$${}_nP_r = \frac{n!}{(n-r)!}$$



Find the number of ways that **6 teachers** can be assigned to **4 sections of an introductory psychology course** if no teacher is assigned to more than one section.

$${}_nP_r = \frac{n!}{(n-r)!}$$

$${}_6P_4 = \frac{6!}{(6-4)!} = 360 \text{ Ans}$$

Counting Sample points

✓ 1. Multiplication Rule

$$= n_1 \times n_2 \times \dots \times n_k$$

✓ 2. Combinations

$${}_n C_r = \frac{n!}{r! (n-r)!}$$

3. Permutations

✓ 3.1 Permutations without constraints

3.2 Permutations with constraints

$${}_n P_r = \frac{n!}{(n-r)!}$$

Counting Sample points

✓ 1. Multiplication Rule

$$= n_1 \times n_2 \times \dots \times n_k$$

✓ 2. Combinations

$${}_n C_r = \frac{n!}{r! (n-r)!}$$

3. Permutations

✓ 3.1 Permutations without constraints

3.2 Permutations with constraints

$${}_n P_r = \frac{n!}{(n-r)!}$$

Permutations with constraints

- Repeated (identical) outcomes

Permutations with constraints

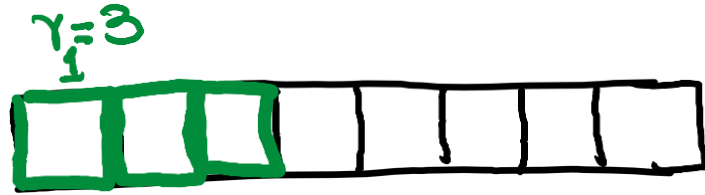
- Repeated (identical) outcomes

$$n=8$$



Permutations with constraints

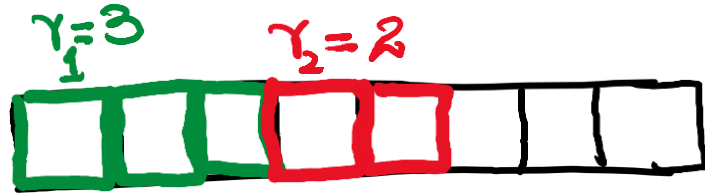
- Repeated (identical) outcomes



$$n=8$$

Permutations with constraints

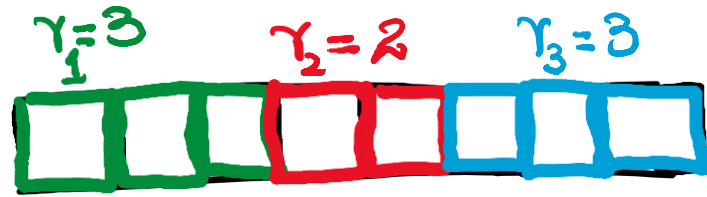
- Repeated (identical) outcomes



$$n=8$$

Permutations with constraints

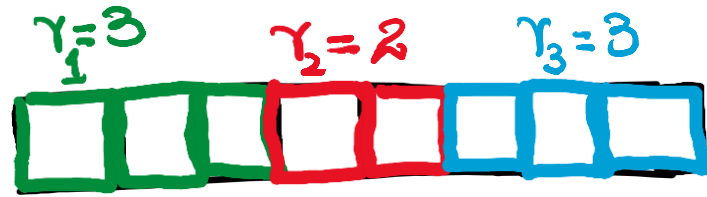
- Repeated (identical) outcomes



$$n = 8$$

Permutations with constraints

- Repeated (identical) outcomes



$$n=8$$

$$\text{Permutations} = \frac{n!}{\gamma_1! \gamma_2! \gamma_3!}$$



How many distinct permutations can be made from the letters of the word INFINITY ?



How many **distinct permutations** can be made from the letters of the word **INFINITY** ?

$$n = 8$$

$$r_1 = 3$$

$$r_2 = 2$$

$$P = \frac{n!}{r_1! r_2!}$$



How many **distinct permutations** can be made from the letters of the word **INFINITY** ?

$$P = \frac{8!}{3!2!} = 3,360 \quad \underline{\text{Ans}}$$

$$n = 8$$

$$r_1 = 3$$

$$r_2 = 2$$

$$P = \frac{n!}{r_1! r_2!}$$



In how many ways can 3 oaks, 4 pines, and 2 maples be arranged along a property line if one does not distinguish among trees of the same kind?



In how many ways can **3 oaks**, **4 pines**, and **2 maples** be arranged along a property line if **one does not distinguish** among trees of the same kind?

$$P = \frac{9!}{3! 4! 2!} = 1,260 \quad \underline{\text{Ans}}$$

Counting Sample points

✓ 1. Multiplication Rule

$$= n_1 \times n_2 \times \dots \times n_k$$

✓ 2. Combinations

$${}_n C_r = \frac{n!}{r! (n-r)!}$$

3. Permutations

✓ 3.1 Permutations without constraints

$${}_n P_r = \frac{n!}{(n-r)!}$$

✓ 3.2 Permutations with constraints

$$P = \frac{n!}{r_1! r_2! r_3! \dots}$$

Counting Sample points

✓ 1. Multiplication Rule

$$= n_1 \times n_2 \times \dots \times n_k$$

✓ 2. Combinations

$${}_n C_r = \frac{n!}{r! (n-r)!}$$

3. Permutations

✓ 3.1 Permutations without constraints

$${}_n P_r = \frac{n!}{(n-r)!}$$

✓ 3.2 Permutations with constraints

$$P = \frac{n!}{r_1! r_2! r_3! \dots}$$

3.3 Permutations alternate outcomes

Permutations with constraints

- alternate outcomes

Permutations with constraints

- alternate outcomes

$$n=7$$



Permutations with constraints

- alternate outcomes

$$n=7$$



Permutations with constraints

- alternate outcomes

$$n=7$$



Permutations with constraints

- alternate outcomes

$$n=7$$



$$P = 4! 3!$$

Permutations with constraints

- alternate outcomes

$$n=7$$



$$P = 4! 3!$$

In General

$$P = n_1! n_2! n_3! \dots$$



In how many ways can 4 boys and 5 girls sit in a row if the boys and girls must alternate?



In how many ways can **4 boys** and **5 girls** sit in a row if the **boys and girls must alternate**?

$$P = 4! 5! = 2,880 \text{ Ans}$$



Four married couples have bought 8 seats in the same row for a concert. In how many different ways can they be seated, if each couple is to sit together?



Four married couples have bought 8 seats in the same row for a concert. In how many different ways can they be seated, if each couple is to sit together?

$$P = 4! 4! = 576 \quad \underline{\text{Ans}}$$

Counting Sample points

✓ 1. Multiplication Rule

$$= n_1 \times n_2 \times \dots \times n_k$$

✓ 2. Combinations

$${}_n C_r = \frac{n!}{r! (n-r)!}$$

3. Permutations

✓ 3.1 Permutations without constraints

$${}_n P_r = \frac{n!}{(n-r)!}$$

✓ 3.2 Permutations with constraints

$$P = \frac{n!}{r_1! r_2! r_3! \dots}$$

✓ 3.3 Permutations alternate outcomes

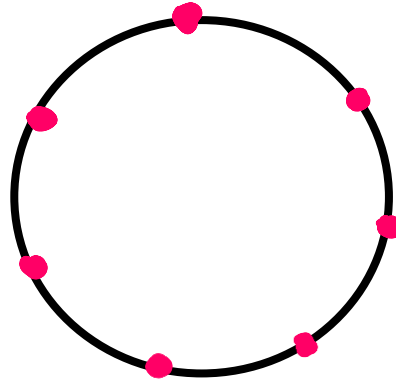
$$P = n_1! n_2! n_3! \dots$$

3.4 Circular Permutations

Circular Permutations

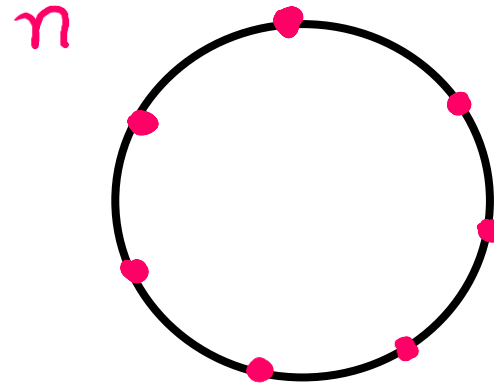
Circular Permutations

n



Circular Permutations

$$P = (n-1)!$$





In how many ways can 5 different trees be planted in a circle?



In how many ways can **5 different trees** be planted **in a circle**?

$$P = (5-1)! = 24 \quad \text{Ans}$$

Counting Sample points

✓ 1. Multiplication Rule

$$= n_1 \times n_2 \times \dots \times n_k$$

✓ 2. Combinations

$${}_n C_r = \frac{n!}{r! (n-r)!}$$

3. Permutations

✓ 3.1 Permutations without constraints

$${}_n P_r = \frac{n!}{(n-r)!}$$

✓ 3.2 Permutations with constraints

$$P = \frac{n!}{r_1! r_2! r_3! \dots}$$

✓ 3.3 Permutations alternate outcomes

$$P = n_1! n_2! n_3! \dots$$

✓ 3.4 Circular Permutations

$$P = (n-1)!$$